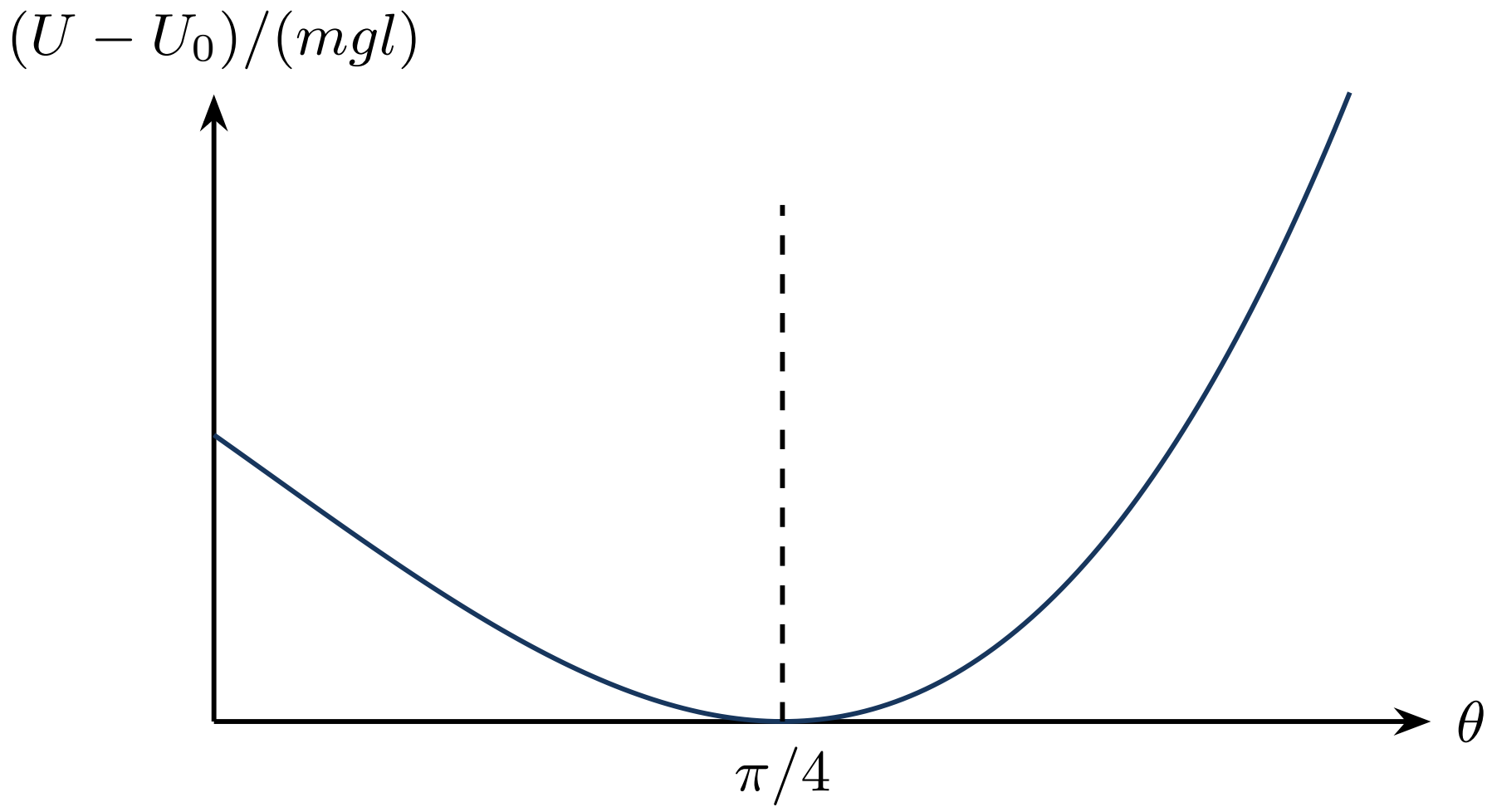

$U''(\pi/4) = 2kl^2 > 0$；图中能量自平衡值起算，横轴范围有限。